

Python Programs: List, Tuple, Set and Dictionary

Beginner-friendly programs with sample output

1. List Program

```
# Create and display a list
fruits = ["Apple", "Banana", "Mango"]

print("Fruits:", fruits)
print("First fruit:", fruits[0])
print("Number of fruits:", len(fruits))
```

Output:

```
Fruits: ['Apple', 'Banana', 'Mango']
First fruit: Apple
Number of fruits: 3
```

2. Tuple Program

```
# Create and display a tuple
numbers = (10, 20, 30, 40)

print("Tuple:", numbers)
print("First number:", numbers[0])
print("Number of elements:", len(numbers))
```

Output:

```
Tuple: (10, 20, 30, 40)
First number: 10
Number of elements: 4
```

3. Set Program

```
# Create and display a set
numbers = {10, 20, 30, 40}

print("Set:", numbers)
numbers.add(50)
print("After adding 50:", numbers)
print("Number of elements:", len(numbers))
```

Output:

```
Set: {10, 20, 30, 40}
After adding 50: {10, 20, 30, 40, 50}
Number of elements: 5
```

4. Dictionary Program

```
# Create and display a dictionary
student = {
    "name": "Ravi",
    "age": 19,
    "course": "AI & DS"
}

print("Student:", student)
print("Name:", student["name"])
print("Course:", student["course"])
```

Output:

```
Student: {'name': 'Ravi', 'age': 19, 'course': 'AI & DS'}
Name: Ravi
Course: AI & DS
```

Note: Set elements may appear in a different order when you run the program because sets are unordered.